

Experiment34:

Consider a file system where the records of the file are stored one after another both physically and logically. A record of the file can only be accessed by reading all the previous records. Design a C program to simulate the file allocation strategy.

Code: -

```
#include <stdio.h>

#include <stdlib.h>

struct Record
{
    int recordNumber;
    char data[256];
};

int main()
{
    FILE *file;
    struct Record record;
    int recordNumber;
    file = fopen("sequential_file.txt", "w");

    if (file == NULL)
    {
        printf("Error opening the file.\n");
        return 1;
    }
```

```
printf("Enter records (Enter '0' as record number to exit):\n");
```

```
while (1)
```

```
{
```

```
    printf("Record Number: ");
```

```
    scanf("%d", &record.recordNumber);
```

```
    if (record.recordNumber == 0)
```

```
    {
```

```
        break;
```

```
    }
```

```
    printf("Data: ");
```

```
    scanf(" %[^\n]", record.data);
```

```
    fwrite(&record, sizeof(struct Record), 1, file);
```

```
}
```

```
fclose(file);
```

```
file = fopen("sequential_file.txt", "r");
```

```
if (file == NULL)
```

```
{
```

```
    printf("Error opening the file.\n");
```

```
    return 1;
```

```
}
```

```
while (1)
{
    printf("\nEnter the record number to read (0 to exit): ");
    scanf("%d", &recordNumber);

    if (recordNumber == 0)
    {
        break;
    }
    while (fread(&record, sizeof(struct Record), 1, file))
    {
        printf("Record Number: %d\n", record.recordNumber);
        printf("Data: %s\n", record.data);

        if (record.recordNumber == recordNumber)
        {
            break;
        }
    }
    rewind(file);
}

fclose(file);
return 0;
}
```

Output:

```
Enter records (Enter '0' as record number to exit):  
Record Number: 0
```

```
Enter the record number to read (0 to exit): 2
```

```
Enter the record number to read (0 to exit): 5
```

```
Enter the record number to read (0 to exit): 7
```

```
Enter the record number to read (0 to exit): 0
```

```
...Program finished with exit code 0
```

```
Press ENTER to exit console.[]
```